Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

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gem5 TimingSimpleCPU · workload profile `train` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, L1D 2-way, L2 8-way · gem5 `e0a12aa4e3+dirty`

Definitions (metric schema 9). Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` $\leq$ `store` and `load` $\geq$ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

Benchmark	ROI_insts	stack_event	**spill_event**	csave_event	other_event	spill/stack %	Spill %	Csave %	L1D m %	L2 m %	sp_load m %	sp_store m %	cs_load m %	cs_store m %
cactuBSSN	383 945 680 437	30 560 138 315	30 377 509 662	181 932 059	696 594	99.4	21.33	0.09	15.768	64.32	4.459	3.897	0.124	0.149
parest	299 200 615 391	580 499 665	94 157 487	462 351 877	23 990 301	16.2	0.18	0.31	11.513	61.90	4.712	2.954	2.226	2.168
deepsjeng	399 987 651 203	14 214 822 378	3 538 346 146	10 408 044 378	268 431 854	24.9	2.99	5.20	9.802	6.58	2.937	7.867	2.873	3.816
leela	436 316 202 851	13 338 709 984	512 185 059	12 317 582 149	508 942 776	3.8	0.43	5.65	6.412	16.65	7.022	0.201	1.846	0.874
namd	218 427 292 248	2 350 200 000	2 339 066 714	8 976 130	2 157 156	99.5	3.22	0.01	12.792	25.42	5.330	1.389	4.247	1.170
lbm	100 160 472 568	4 229 647 076	4 229 634 900	11 332	844	100.0	9.69	0.00	23.364	88.88	0.424	2.108	10.784	0.891
nab	38 232 521 997	784 372 830	637 890 733	128 504 979	17 977 118	81.3	7.61	0.67	7.239	4.20	0.995	0.807	1.159	1.446
nab_gcn4dna	366 315 945 523	7 770 025 698	5 788 787 408	1 746 098 454	235 139 836	74.5	7.30	0.95	6.931	41.67	1.143	1.378	1.260	1.841
xz	45 071 901 345	1 313 479 362	149 252 420	1 077 227 617	86 999 325	11.4	1.36	4.78	7.110	32.18	1.406	9.928	0.562	2.003
xz_combined	103 539 847 907	1 438 295 106	686 344 414	673 142 677	78 808 015	47.7	2.79	1.30	10.851	20.84	3.701	2.298	3.716	4.200
imagick	284 114 941 059	294 947 574	3 377 502	291 114 095	455 977	1.1	0.04	0.20	5.021	94.86	34.988	0.470	0.076	0.042
blender	721 231 649 259	11 980 515 650	494 531 875	10 340 415 008	1 145 568 767	4.1	0.31	2.87	7.420	11.64	3.841	7.013	3.816	3.949
povray	28 621 075 427	2 169 255 652	27 550 868	2 002 183 119	139 521 665	1.3	0.27	13.99	10.102	41.87	7.634	20.369	1.987	2.166
omnetpp	134 781 952 422	8 710 005 757	5 235 206	8 476 934 143	227 836 408	0.1	0.01	12.58	16.549	28.83	0.250	0.873	3.586	3.090
xalancbmk	270 346 763 301	25 872 371 471	370 722 439	24 133 124 322	1 368 524 710	1.4	0.44	17.85	12.401	12.36	20.478	29.684	2.009	3.442
x264	386 397 511 704	16 699 463 061	11 425 519 865	4 723 224 155	550 719 041	68.4	6.30	2.44	2.327	28.09	0.391	0.945	1.291	1.441
gcc_200	118 445 469 526	9 732 539 869	435 131 547	9 187 999 599	109 408 723	4.5	1.16	15.51	9.558	36.44	2.260	5.111	0.947	1.775
gcc_scilab	92 316 903 709	7 134 405 901	365 548 253	6 690 997 652	77 859 996	5.1	1.25	14.50	10.166	36.72	2.616	4.684	0.943	1.860
gcc	10 027 112 139	797 875 479	42 973 169	745 789 454	9 112 856	5.4	1.41	14.88	9.938	38.95	2.414	4.845	0.924	1.921
mcf	125 387 999 079	1 200 460 284	483 048 594	714 151 251	3 260 439	40.2	1.61	1.14	23.825	75.08	7.037	8.489	3.741	5.922
perlbench_scrabbl	29 794 284 033	2 637 314 691	19 766 463	2 595 400 347	22 147 881	0.7	0.23	17.42	7.971	11.62	3.139	11.629	1.532	1.850
perlbench_suns	1 489 955 308	141 093 681	4 683 473	133 092 545	3 317 663	3.3	0.98	17.87	6.547	18.32	1.109	8.999	0.638	1.128
perlbench_diffmail	71 841 738 045	5 980 785 368	357 324 444	5 559 068 861	64 392 063	6.0	2.06	15.48	6.459	32.33	4.867	4.168	0.433	1.420
perlbench_perfect	12 407 457 725	1 157 150 004	57 361 433	1 079 863 927	19 924 644	5.0	1.44	17.41	9.270	51.16	1.241	5.864	0.658	1.422
perlbench_splitmail	36 465 452 492	2 676 936 460	458 223 604	2 146 373 965	72 338 891	17.1	5.79	11.77	5.379	17.25	1.788	2.776	1.168	0.882
bwaves_1	94 269 943 053	669 560 992	669 492 761	55 690	12 541	100.0	2.99	0.00	42.240	27.05	3.276	0.266	4.574	5.163
bwaves_2	91 219 148 006	659 168 858	659 100 916	55 475	12 467	100.0	3.04	0.00	42.111	27.48	3.213	0.270	4.496	5.146
exchange2	177 369 845 254	5 445 875 346	3 838 435 663	1 053 305 830	554 133 853	70.5	4.84	1.19	2.265	1.44	2.049	2.700	5.466	4.162
fotonik3d	188 709 212 304	1 778 874 318	1 763 902 508	13 601 309	1 370 501	99.2	3.99	0.01	17.946	38.50	5.705	0.057	0.155	3.923
wrf	519 187 619 962	2 589 057 857	681 696 831	1 750 897 417	156 463 609	26.3	0.92	0.67	17.589	22.33	9.173	6.100	0.861	1.685
cam4	63 971 148 446	1 115 718 661	416 171 917	673 858 888	25 687 856	37.3	2.49	2.11	11.199	54.29	2.588	2.625	1.073	1.861
Total	5 849 595 313 723	186 023 567 348	**70 932 974 274**	109 315 378 704	5 775 214 370	38.1	3.60	3.74	0.000	—	—	—	—	—